

Quiz: #03

Name:

ID:

Section:

Question 01: You are a satellite engineer at a deep-space communication center. Write a Java program that reads a single line of input containing both uppercase and lowercase letters, removes all + and = characters, and calculates the Total Signal Strength using the formula **(total number of letters in the entire cleaned sentence) × (total number of + and = removed from the entire sentence)**. Spaces are not counted as letters. At the end, print the cleaned sentence and the Total Signal Strength. You are not allowed to use any built-in functions except `charAt()`, `length()`, and `equals()`. (7 marks)

Sample Input	Sample Output
Enter planet codes: aL+ph=a Be+ta G+a+m=ma	Cleaned Sentence: aLpha Beta Gamma Total Signal Strength: 210 Explanation: After removing all + and = from the entire sentence and ignoring spaces, the total number of letters counted is 14, and the total number of + and = removed is 6, so the Total Signal Strength = $14 \times 6 = 210$.

Question 01: Write a method named **EdgeDigitProcessor** that takes an integer array as input. For each number in the array, if the sum of its first and last digits is greater than 10, compute the square of the number and add it to a total called **highSum**; otherwise, compute the product of its first and last digits and add it to another total called **lowSum**. After processing all elements, calculate the Final Edge Score using the formula: **(highSum) + (lowSum × 3) + total number of elements**, and return this value. (7 marks)

Sample Method Call	Sample Output
<pre>int ans = EdgeDigitProcessor(new int[]{45, 123, 89, 210, 67}); System.out.println(ans);</pre>	12484 Explanation: In the array {45, 123, 89, 210, 67}, numbers whose sum of first and last digits is greater than 10 (89, 67) are squared and added to highSum, while the others (45, 123, 210) contribute the product of their first and last digits to lowSum. Finally, the Final Edge Score is calculated using the formula: (highSum) + (3 × lowSum) + total number of elements.

Dept. of Computer Science & Engineering, BRAC University
 Quiz #04
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11/11/2019 - Spring 2019

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Illustrate the outputs of the following statements, provide your solution in the answer script to verify your outputs. All outputs must appear in the answer script paper,

```

1 class TracerB {
2     public static void main(String args[]) {
3         int[] myArray = new int[5];
4         int index1 = 0, index2 = 0;
5         while (index1 < 3) {
6             myArray[index1] = (index1 + 2) * 2;
7             index2 = 1;
8             while (index2 < index1) {
9                 myArray[index1] = myArray[index1] + myArray[index2] / index2;
10                index2 = index2 + 1;
11            }
12            if (index1 <= 4) {
13                System.out.println(myArray[index1]);
14            }
15            index1 = index1 + 1;
16        }
17    }
18 }
  
```

Output